

(a) Arrival

electronics package, one of three (resistojets, solar cells)

Kevlar lines to nose and tail hold the rod's angle; 8 m long, not to scale

66–75 km/s

spin 29°/s

solid polyethylene rod, 2.5 kg, 0.69 m by 6.9 cm, metal under a clear coat, no electronics

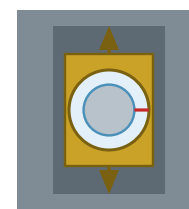
ship's forward face, standoff foil dashed

sliding door carries the port and plug, trims to the tracked intercept, then locks (door drawn enlarged)

methane charge

polyethylene foam plug in a solid skin, 0.98 m deep, 10.9 cm wide, 4.4 kg

exhaust throat, 0.44–0.50 m across, 16–21× entrance area



door slides in two axes
port 10.9 cm
rod 6.9 cm
2 cm clear

(b) After impact (sketch)

tether lines strike the standoff foil; packages pass outside the ship

a small share leaks back out the 10.9 cm port

front spreads as it sweeps the charge; pressure clamps the door to its backing

rod and plug turn to plasma together

most gas leaves through the throat

